

Fundamentals of Information Systems Security

Chapter 2 Quiz: Emerging Technologies

Name: _____

Date: _____

Total Points: 100

Time Allowed: 30 Minutes

Instructions

- This quiz contains **20 questions** covering Chapter 2
- Answer all questions to the best of your ability
- Multiple choice questions have only one correct answer
- True/False questions should be marked T or F
- Short answer questions require concise, accurate responses

Section A: Multiple Choice (40 points)

1. (4 points) Who first coined the term "Internet of Things" (IoT) in 1999?
 - A. Bill Gates
 - B. Steve Jobs
 - C. Kevin Ashton
 - D. Tim Berners-Lee
2. (4 points) Which technology was a precursor to IoT and used for tracking goods in supply chains?
 - A. GPS
 - B. RFID
 - C. NFC
 - D. Bluetooth
3. (4 points) Which of the following is NOT listed as a key driver for IoT growth?
 - A. Global IP-based networking
 - B. Smaller and faster computing
 - C. Growth of cloud computing
 - D. Decreasing internet speed
4. (4 points) What percentage of email traffic is estimated to be spam?
 - A. 30-50%
 - B. 50-70%
 - C. 70-90%

- D. 90-98%
5. (4 points) Which of the following is NOT one of the five critical challenges of IoT?
- A. Security
 - B. Privacy
 - C. Interoperability
 - D. Cost reduction
6. (4 points) What is the IoT challenge that deals with devices from different manufacturers communicating seamlessly?
- A. Security
 - B. Privacy
 - C. Interoperability
 - D. Regulatory compliance
7. (4 points) Which mode of communication is used for urgent matters requiring an immediate response?
- A. Store-and-forward communication
 - B. Batch processing
 - C. Real-time communication
 - D. Asynchronous communication
8. (4 points) Which technology involves manipulating matter at the atomic and molecular scale (1-100 nanometers)?
- A. 3D Printing
 - B. Artificial Intelligence
 - C. Nanotechnology
 - D. Robotics
9. (4 points) What is the primary security concern regarding IoT devices?
- A. They are too expensive to secure
 - B. They often have limited processing power and security features
 - C. They cannot connect to the internet
 - D. They are not used by organizations
10. (4 points) Which of the following is an example of how IoT impacts human life?
- A. Health monitoring wearables
 - B. Manual inventory tracking
 - C. Paper-based record keeping
 - D. Traditional landline telephones

Section B: True/False (20 points)

11. (2 points) Information security performs four important functions including protecting the organization's ability to function.
 - A. True
 - B. False
12. (2 points) 3D printing promises to disrupt traditional manufacturing by enabling mass customization.
 - A. True
 - B. False
13. (2 points) The convergence of emerging technologies does not amplify their individual impact.
 - A. True
 - B. False
14. (2 points) IoT amplifies existing internet challenges, introducing new layers of complexity in security and privacy.
 - A. True
 - B. False
15. (2 points) The law always keeps pace with rapid technological change.
 - A. True
 - B. False
16. (2 points) B2C stands for Business-to-Consumer in the context of digital business models.
 - A. True
 - B. False
17. (2 points) Smart cities use IoT for traffic monitoring and management.
 - A. True
 - B. False
18. (2 points) Privacy is not a concern with IoT devices since they only collect anonymous data.
 - A. True
 - B. False
19. (2 points) The evolution from RFID to IoT represents a shift from simple tracking to complex ecosystems of intelligent, data-sharing devices.
 - A. True
 - B. False
20. (2 points) Telemedicine and remote surgery are not enabled by emerging technologies.
 - A. True
 - B. False

Section C: Short Answer (40 points)

21. (10 points) List and describe the five key drivers for IoT growth discussed in Chapter 2.
22. (10 points) Explain the five critical challenges that must be addressed for the safe and effective adoption of IoT.
23. (10 points) Describe how IoT impacts both human life and business operations. Provide at least two examples for each.
24. (10 points) What is the critical note about interoperability failures and security vulnerabilities in IoT?

End of Quiz